MEME16203Linear Models

Assignment 2

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Due by:			

Q1. A food scientist performed the following experiment to study the effects of combining two different fats and two different surfactants on the specific volume of bread loaves. Three batches of dough were made for each of the four combinations of fat and surfactant. Ten loaves of bread were made from each batch of dough and the average volume of the ten loaves was recorded for each batch. The data are as below.

	Surfactant	
	A	B
Fat 1	6.7	7.1
	4.3	5.9
	5.7	5.6
Fat 2	5.9	5.6
	7.4	6.8
	7.1	6.5

Consider the model $Y_{ijk} = \mu + \alpha_i + \beta_j + \gamma_{ij} + \epsilon_{ijk}$ where $\epsilon_{ijk} \sim NID(0, \sigma^s 2)$ and Y_{ijk} denotes the average of the volumes of ten loaves of bread made from the k^{th} batch of dough using the i^{th} fat and the j^{th} surfactant. Consider

$$\boldsymbol{\beta} = \begin{bmatrix}\mu \ \alpha_1 \ \alpha_2 \ \beta_1 \ \beta_2 \ \gamma_{11} \ \gamma_{12} \ \gamma_{21} \ \gamma_{22}\end{bmatrix}^T.$$

For each of the following linear functions of the parameters, determine if it is estimable. If it is estimable, give a vector $\mathbf{a}$ such that $E(\mathbf{a^T Y})$ is equal to that linear combination of parameters. If it is estimable, describe what that linear combination of parameters represents with respect to the effects of the fats and surfactants on mean bread volume.

(i)μ

(ii)α_2

(iii)$\beta_1 - \beta_2$

(iv)γ_{22}

(v)$\mu + \alpha_2 + \beta_2 + \gamma_{22}$

(vi)$\gamma_{11} - \gamma_{12}$

(vii)$\gamma_{11} - \gamma_{12} - \gamma_{21} + \gamma_{22}$

(viii)$(\beta_1 - \beta_2) + \frac{1}{2}(\gamma_{11} + \gamma_{21} - \gamma_{12} - \gamma_{22})$

(20 marks)

Q2. Suppose $y_i = \beta_0 + \beta_1 x_i + \epsilon_i$, where for $\boldsymbol{\epsilon} = (\epsilon_1, \epsilon_2, \cdots, \epsilon_n)'$, $E(\boldsymbol{\epsilon}) = \mathbf{0}$. A particular experiment produces $n = 5$ data points as per

x_i	28	36	35	38	92
y_i	331	387	110	140	83

Suppose that $V(\boldsymbol{\epsilon}) = \sigma^2 diag\left(\frac{1}{4}, \frac{1}{16}, \frac{1}{49}, \frac{1}{81}, \frac{1}{144}\right)$. Evaluate an appropriate BLUE of $\boldsymbol{\beta} = \begin{bmatrix} \beta_0 \\ \beta_1 \end{bmatrix}$ under the model assumptions. (20marks)

Q3. Two varieties of palm oil tree(variety I and variety II) were compared in a field trail. In addition to the varieties, three levels of nitrogen were used (45, 95, and 145 kilograms per acre (kg/a). Six different fields were used, and the six combinations of varieties and nitrogen levels were randomly assigned to the fields. Let Y_{ij} denote the yield (in bushels per acre) of the i^{th} variety of palm oil tree when the j^{th} level of nitrogen is applied. Throughout this question, ϵ_{ij}, $i = 1, 2, j = 1, 2, 3$, denote independent $N(0, \sigma^2)$ random variables where σ^2 is an unknown variance. The following two models were proposed:

$$\text{Model 1:} \quad \begin{bmatrix} Y_{11} \\ Y_{12} \\ Y_{13} \\ Y_{21} \\ Y_{22} \\ Y_{23} \end{bmatrix} = \begin{bmatrix} 1 & 0 & -50 & 2500 \\ 1 & 0 & 0 & 0 \\ 1 & 0 & 50 & 2500 \\ 0 & 1 & -50 & 2500 \\ 0 & 1 & 0 & 0 \\ 0 & 1 & 50 & 2500 \end{bmatrix} \begin{bmatrix} \theta_1 \\ \theta_2 \\ \rho_1 \\ \rho_2 \end{bmatrix} + \begin{bmatrix} \epsilon_{11} \\ \epsilon_{12} \\ \epsilon_{13} \\ \epsilon_{21} \\ \epsilon_{22} \\ \epsilon_{23} \end{bmatrix}$$

$$\text{Model 2:} \quad \begin{bmatrix} Y_{11} \\ Y_{12} \\ Y_{13} \\ Y_{21} \\ Y_{22} \\ Y_{23} \end{bmatrix} = \begin{bmatrix} 1 & 1 & 0 & -1 & 1 \\ 1 & 1 & 0 & 0 & -2 \\ 1 & 1 & 0 & 1 & 1 \\ 1 & 0 & 1 & -1 & 1 \\ 1 & 0 & 1 & 0 & -2 \\ 1 & 0 & 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} \mu \\ \alpha_1 \\ \alpha_2 \\ \beta_1 \\ \beta_2 \end{bmatrix} + \begin{bmatrix} \epsilon_{11} \\ \epsilon_{12} \\ \epsilon_{13} \\ \epsilon_{21} \\ \epsilon_{22} \\ \epsilon_{23} \end{bmatrix}$$

(a) With respect to the effects of varieties and nitrogen levels on corn yields, interpret the parameters θ_2 and ρ_2 in Model 1.

(b) For Model 1, indicate which of the following quantities are estimable

$$\theta_1 - \theta_2; \qquad \theta_1 - 10\rho_1 + 100\rho_2$$

Give a brief explanation, to support your conclusions.

(c) For Model 2, determine if $\mu + \alpha_2$ is estimable? Give a brief explanation to support your conclusion.

(d) Expressing Model 2 as $\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}$, compute a solution to the normal equations, $\mathbf{b} = (\mathbf{X^T X})^{-}\mathbf{X^T Y}$ in terms of Y_{ij}.

(e) Using $\mathbf{b} = (\mathbf{X^T X})^{-}\mathbf{X^T Y}$ from Part (d), find the unique estimator of $2\mu + \alpha_1 + \alpha_2 + 2\beta_1 + 3\beta_2$.

(f) Show that Model 1 is the reparametrization of Model 2.

(g) Using $\mathbf{b} = (\mathbf{X^T X})^{-}\mathbf{X^T Y}$ from Part (d) and the fact that Model 1 is the reparametrization of Model 2, find the estimator of $\boldsymbol{\gamma}$.

(20 marks)

Q4. Two varieties of corn (variety A and variety B) were compared in a field trail. In addition to the varieties, four levels of nitrogen were used [0, 20, 40 and 60 pounds per acre (lb/a)]. Eight different fields were used, and the eight combinations of varieties and nitrogen levels were randomly assigned to the fields. Consider a Gauss-Markov model

$$y_{ij} = \mu + \alpha_i + \gamma X_j + \epsilon_{ij} \qquad \text{Model (1)}$$

where

- y_{ij} is the yield (in bushels per acre) of the i^{th} variety of corn when the j^{th} level of nitrogen is applied.
- X_j denote the level of nitrogen administered to the corn,
- $\mu, \alpha_1, \alpha_2, \gamma$ are unknown parameters, and
- ϵ_{ij} denotes a random error with $\epsilon_{ij} \sim NID(0, \sigma^2)$ where $\sigma^2 > 0$.

Use this model to answer the following questions. (You may express your answers in matrix notation, but define any new notation that you introduce.)

(a) Let $\boldsymbol{\beta} = (\mu, \alpha_1, \alpha_2, \gamma)^T$, $\mathbf{y} = [y_{11}, y_{12}, y_{13}, y_{14}, y_{21}, y_{22}, y_{23}, y_{24}]^T$, and $\boldsymbol{\epsilon} = [\epsilon_{11}, \epsilon_{12}, \epsilon_{13}, \epsilon_{14}, \epsilon_{21}, \epsilon_{22}, \epsilon_{23}, \epsilon_{24}]^T$. Model (1) can be expressed in the form $\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}$. Do not impose any restriction on the parameters, write down matrix of $\mathbf{X}$ in Kronecker form.

(b) Show that $\mu + \alpha_i + \gamma x$ is estimable for any $x \in \mathbf{R}$.

(c) Show that $\mathbf{X}$ in part (a) has the same column space as

$$\mathbf{W} = \begin{bmatrix} 1 & -1 & -3 \\ 1 & -1 & -1 \\ 1 & -1 & 1 \\ 1 & -1 & 3 \\ 1 & 1 & -3 \\ 1 & 1 & -1 \\ 1 & 1 & 1 \\ 1 & 1 & 3 \end{bmatrix}.$$

(d) Suppose $\mathbf{y} = [7, 8, 4, 11, 4, 6, 9, 6]^T$. Determine the ordinary least squares estimator of $\mu + \alpha_1 + \gamma x$ for any nitrogen levels.

(20 marks)

Q5. In an experiment to investigate the effect of color paper (yellow, green) on response rates for questionnaires distributed by the "windsheield method" in supermarket parking lots, 10 representative supermarket parking lots were chosen in a metropolitan area and each color was assigned random to five of the lots. The reponse rates (in percent) follow.

		j				
	i	1	2	3	4	5
1	Yellow	28	26	31	27	35
2	Green	24	29	25	31	29

Consider the linear model

$$y_{ij} = \mu + \tau_i + \epsilon_{ij}, \text{ for } i = 1, 2 \text{ and } j = 1, 2, \ldots, 5$$

where

- y_{ij} is the observed response rate for the i^{th} color paper assigned to the j^{th} parking lot.
- τ_i corresponds to the effect of i^{th} color.
- $\epsilon_{ij} \sim N(0, \sigma^2)$.

(a) Write model above in the form $\mathbf{y} = X\boldsymbol{\beta} + \boldsymbol{\epsilon}$. Do not impose any restriction on the parameters.

(b) Obtain one of the generalized inverse of $\mathbf{X^T X}$, $\mathbf{G}$.

(c) Use the generalized inverse you obtained in part(b) to compute a solution to the normal equations, $\hat{\boldsymbol{\beta}} = \begin{bmatrix} \hat{\mu} \\ \hat{\tau}_1 \\ \hat{\tau}_2 \end{bmatrix}$.

(d) Using your solution $\hat{\boldsymbol{\beta}}$ to the normal equation from part (c), estimates $\tau_1 - \tau_2$.

(e) Is your estimate in part(d) the unique BLUE? Explain.

(20 marks)